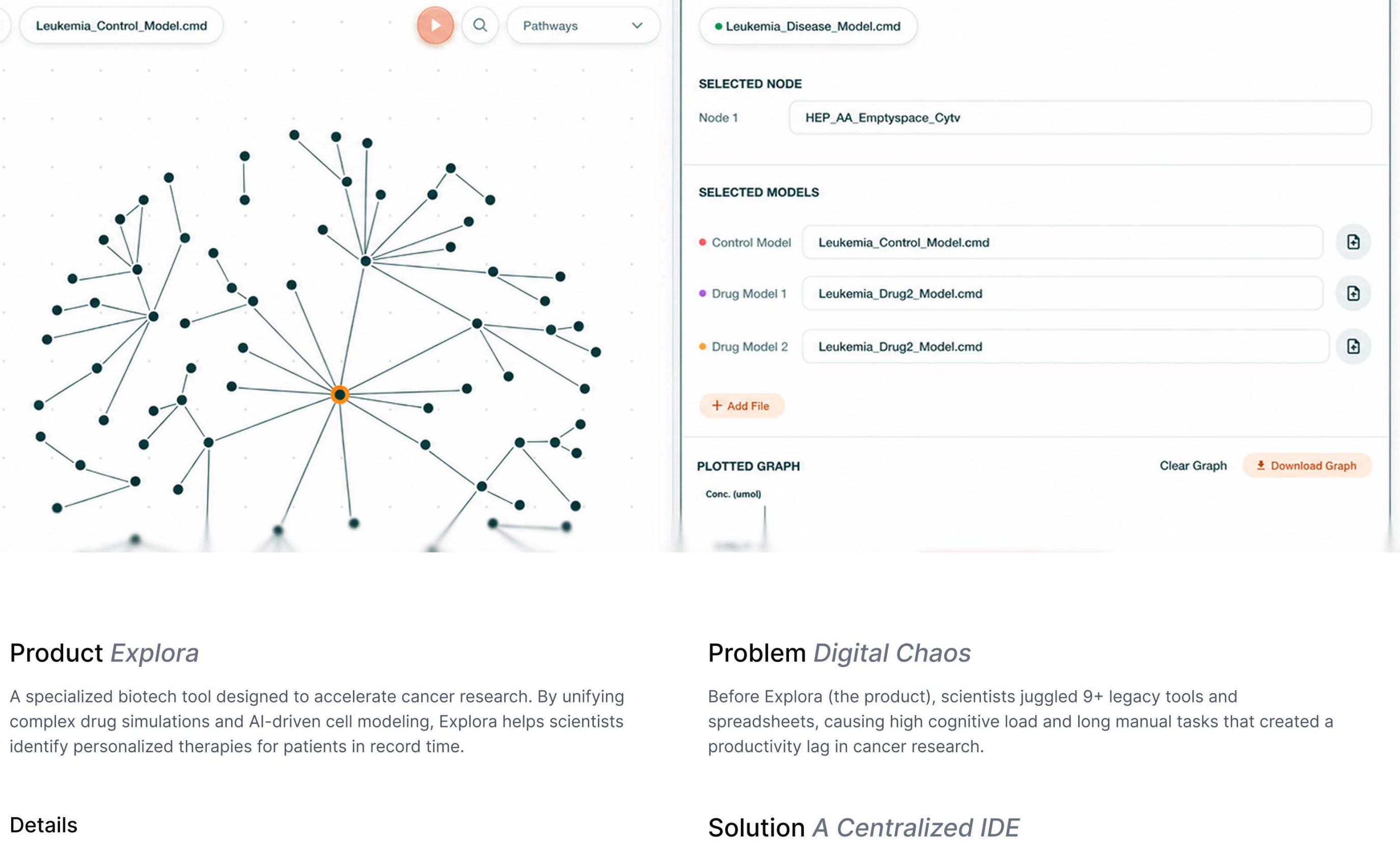


Empowering scientists to deliver faster personalized cancer care

0 to 1 Product Design • Design Systems • Shipped



Product Explora

A specialized biotech tool designed to accelerate cancer research. By unifying complex drug simulations and AI-driven cell modeling, Explora helps scientists identify personalized therapies for patients in record time.

Details

year 2026
type of product Web-based tool, IDE
my role Principal Product Designer (individual designer)
team members CEO - 1 Product Manager - 2 Developers
scope 0 to 1 Product Strategy - UX/UI - Design System
tools Figma - Figma Make - Claude - ChatGPT - Stitch - Weave

Problem Digital Chaos

Before Explora (the product), scientists juggled 9+ legacy tools and spreadsheets, causing high cognitive load and long manual tasks that created a productivity lag in cancer research.

Solution A Centralized IDE

I led the 0 to 1 design to unify fractured systems into a singular IDE, enabling scientists to conduct research within one platform.

Result Productivity Gains

I established the company's first design system of ~80+ components. Product's first launch ensured 100% organization-wide adoption, reducing task time by 12.5% and saving nearly ~\$1.2M in annual productivity costs.

Impact

~\$1.2M

Recovered in Productivity
Based on ~1 hr/day saved per user

100%

Organization Wide Adoption
From C-Suite to Research Scientists

9→1

Systems Consolidated
Unified legacy chaos into a single IDE

CONTEXT

Cellworks is a *biotech company* focused on *curing cancer*.

and how do they do that? They built a machine learning model that simulates human cell behavior. By running FDA-approved drug simulations on it, they identify the most effective and personalized therapies for cancer patients' unique DNA.

Scientists worked on the bio-model to update it as per new research findings and engineers worked on refining the model to produce more accurate results.



GOALS + NORTH STAR

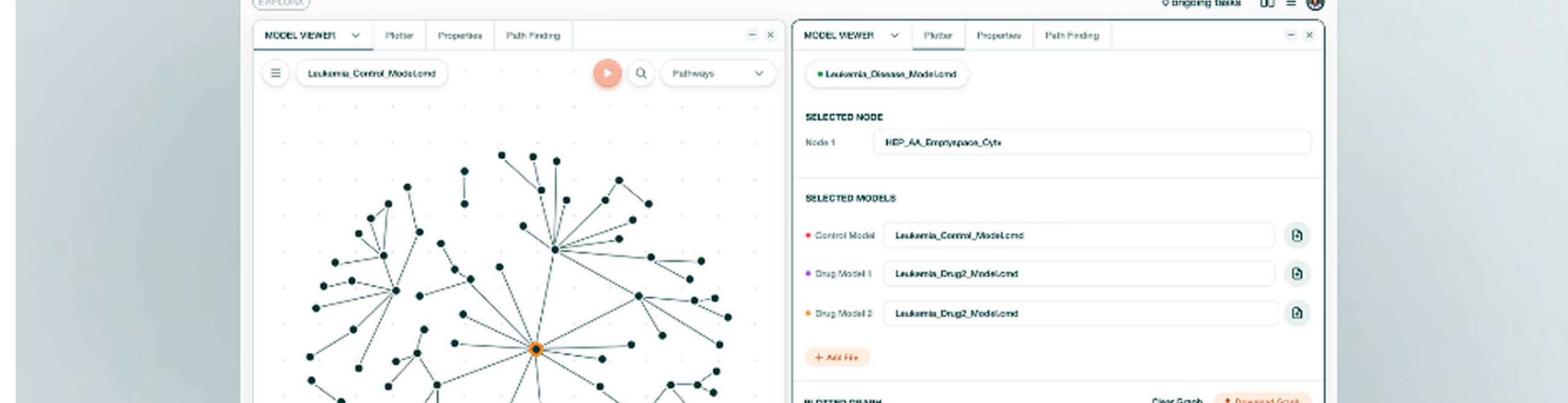
To design a *unified R&D platform*, from the ground up to *accelerate the delivery* of personalized cancer care.

Unify Systems: Consolidate 12+ legacy tools into one cohesive R&D platform to simplify complex workflows.

Accelerate Research: Speed up cancer research by eliminating "tool-hopping" and reducing cognitive strain for scientists.

Increase Throughput: Enable parallel experiments to fully leverage the platform's new asynchronous processing capabilities.

Scale Growth: Establish a foundational design system to efficiently support future biotech applications and features.



PROCESS + KEY INSIGHTS

I conducted contextual inquiries and *interviews* with computational *biologists*, machine learning *engineers*, and leadership to *map disjointed R&D workflows* and *visualize a unified research ecosystem*.

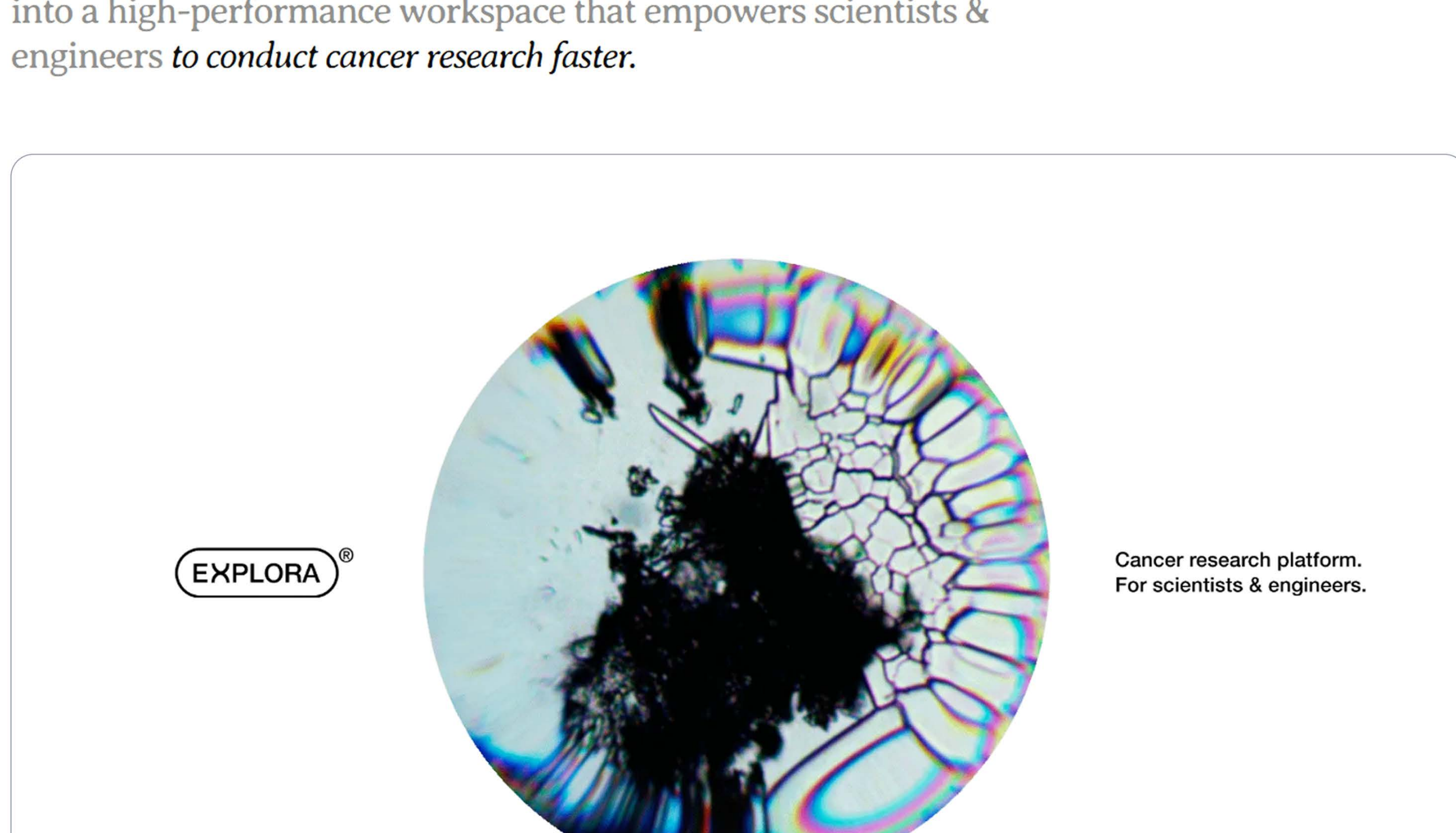
Key Insight 1: Fragmented Ecosystem navigating a scattered ecosystem of 12+ legacy tools forced constant app-switching, which disrupted scientific focus and slowed research progress.

Key Insight 2: Processing Bottleneck the legacy "one-task-at-a-time" system created a gap where expert staff had to sit idle while waiting for experiments to process, losing valuable research time.

Key Insight 3: Manual Insight Cycle relying on Excel exports and external tools to render data created a heavy lag between running an experiment and actually understanding the results.

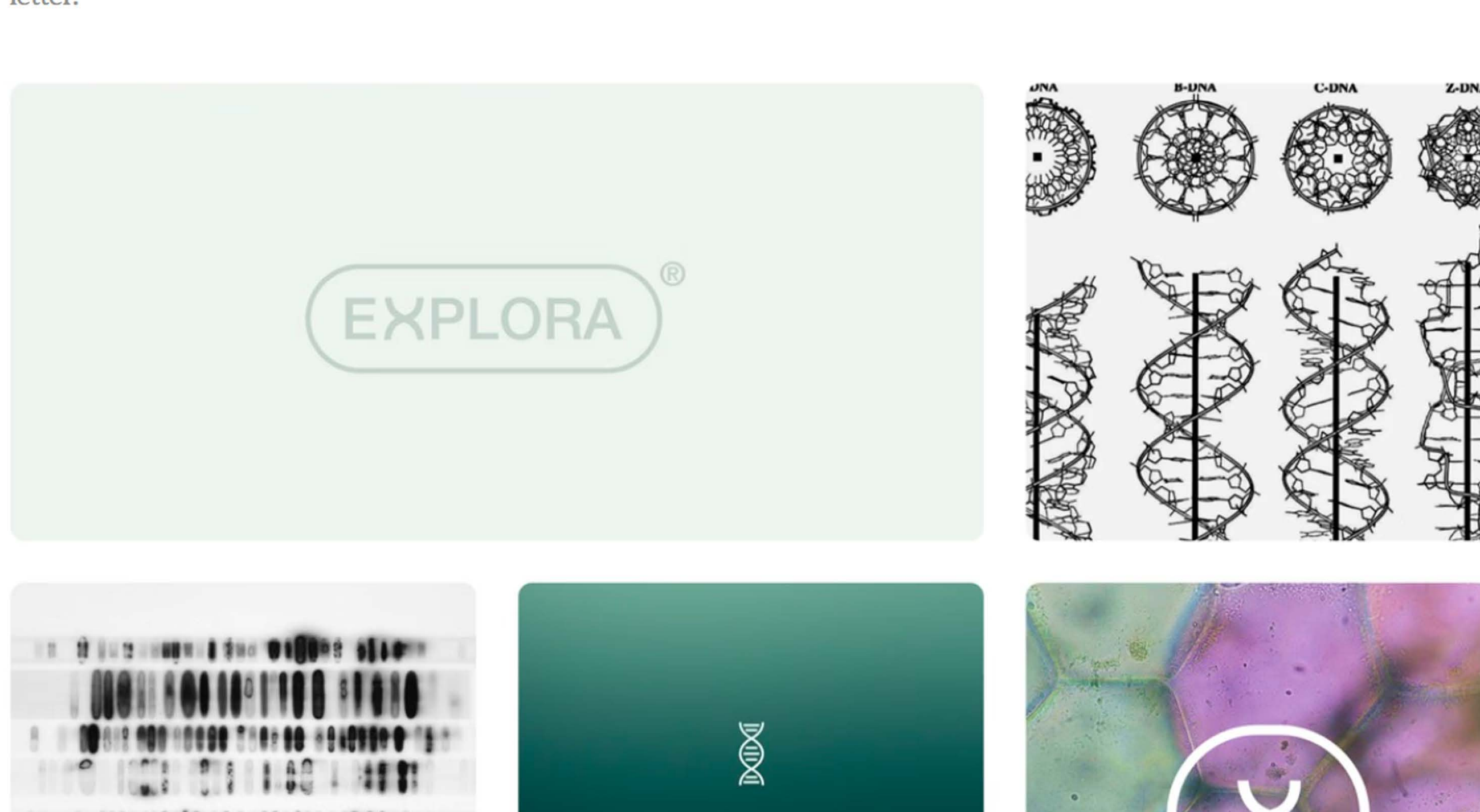
SOLUTION

Explora, a *scalable, systems-driven MVP* consolidating legacy tools into a high-performance workspace that empowers scientists & engineers to *conduct cancer research faster*.



BRANDING

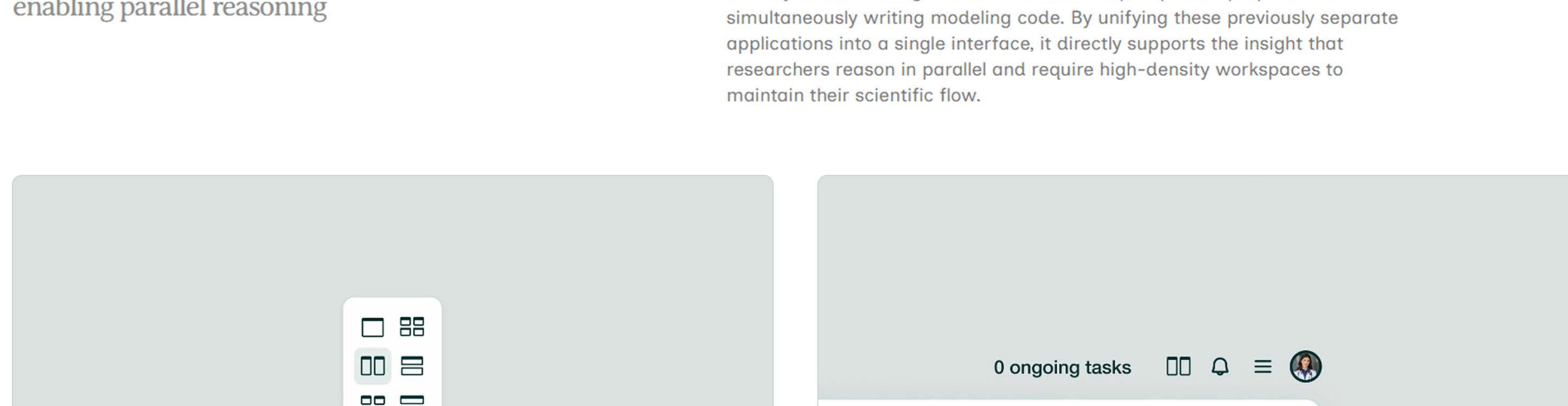
Logo inspired by the *intertwining of the DNA strands*, the unique makeup of a human. The 'X' in 'Explora' replaces the letter.



DESIGN 1/3

The Split-Workspace Grid
enabling parallel reasoning

This feature allows scientists to position the model viewer and code editor side-by-side, enabling them to examine complex protein properties while simultaneously writing modeling code. By unifying these previously separate applications into a single interface, it directly supports the insight that researchers reason in parallel and require high-density workspaces to maintain their scientific flow.



DESIGN 2/3

The Plotter
accelerating insights by visual synthesis

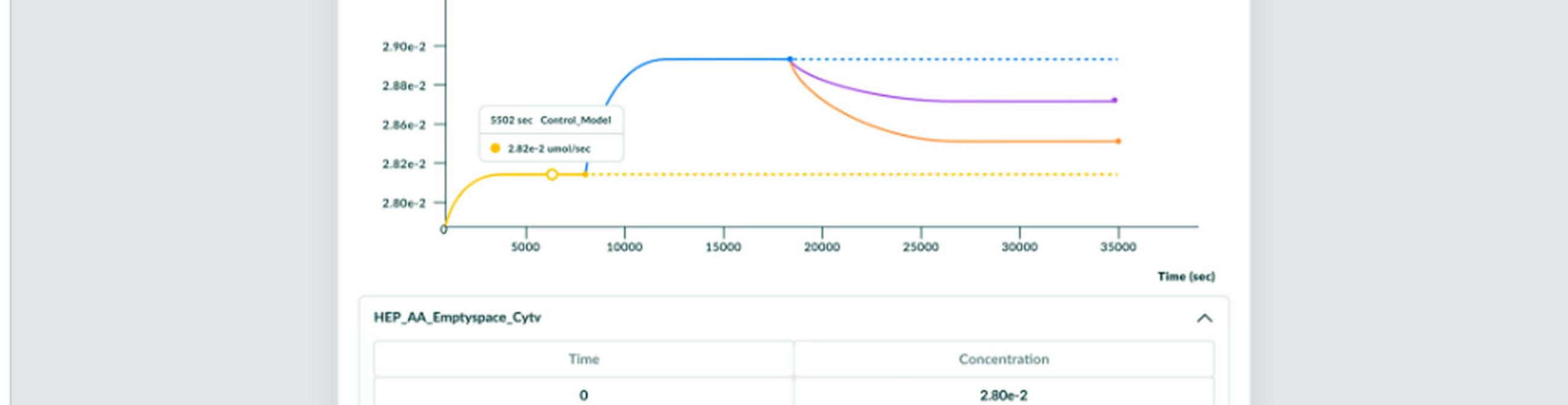
The Plotter allows researchers to load multiple simulation models and visually overlay results in a single, integrated environment. By replacing cognitively heavy Excel exports with at-a-glance comparisons, it solves the "Manual Data Foraging" insight and empowers scientists to iterate on drug developments faster.



DESIGN 3/3

The Progress Bar & View Selector
eliminating research downtime

I designed a system to handle the transition from a synchronous to an asynchronous back-end. While the *View Selector* allows scientists to switch fluidly between active applications during a simulation, the *Ongoing Tasks* panel provides a global, bird's-eye view of experiment progress. Together, these features transformed a legacy 'one-task-at-a-time' bottleneck into a high-throughput environment where scientists can simulate and monitor multiple experiments in parallel.



A scientist always has a bird's view of the progress of his experiment simulations.

A scientist can write a drug experiment code while looking at the model to run it on.

REFLECTIONS

Need for a Deep Domain Understanding: To design for experts, you have to understand their world. By diving into bio-modeling science, I moved from just "translating requirements" to becoming a co-innovator who made complex data more actionable.

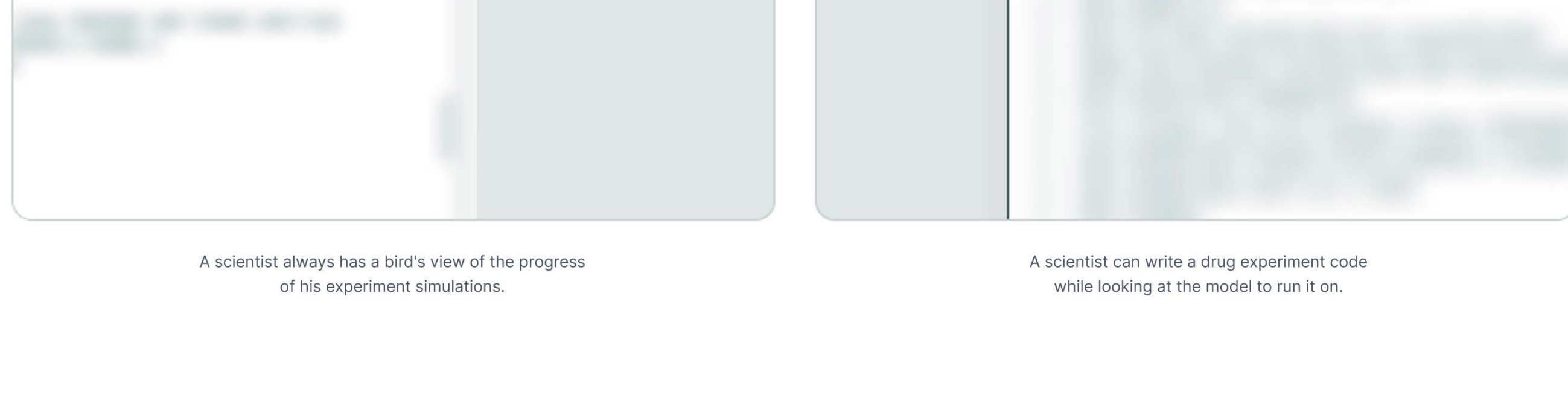
Learning the Art of Prioritization: Working as a solo designer taught me how to prioritize. I built a lean, 8B-component foundation instead of a bloated system, allowing us to hit a tight launch deadline while keeping the platform scalable.

Designing to reduce Cognitive Load: Combining fragmented tools was about psychology, not just tech. By creating a "one-stop" workspace, I reduced the cognitive strain of tool-hopping, helping scientists stay focused on their research instead of their software.

WHAT I WOULD DO NEXT...

Introduce an *AI-powered chat assistant*

Biotech is one of the fastest growing areas where AI is being used extensively for research. Leveraging it, I'd focus on introducing an AI-powered chat assistant primarily focused on automating many of the daily tasks.



Experiment a little with the *homepage design*

Here is a conceptual user interface that I designed as an answer to the question: what would a conversational homepage look like?



Explora's smart scientists and engineers lived happily every after

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